Pre-gravity6plane: the Pin(4,4) fermion field dirac16complex, its 6-planes, gravity, Kohn-Sham pair creation and Einstein-Rosen bridges

This repository extends the Mathematica notebook Pre-gravity6plane.nb (P. L. Nash, (C) 2022, GNU General Public License) with a second-quantized fermion field, dirac16complex, and the calculations requested for it. Every equation and result stated below is computed with exact arithmetic (or a convergent numerical method where stated) in the notebook 6planes.nb.

1. Setting

Coordinates $x_0, \ldots, x_7$ (x_0 hidden space, $x_{1,2,3}$ space, x_4 time, $x_{5,6,7}$ superluminal time), metric $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$. The eight real 16×16 Dirac matrices are those of Pre-gravity6plane.nb,

$$T^A = (T16^A)[A] = \begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}, \qquad T^A T^B + T^B T^A = 2\eta^{AB} I_{16}, \qquad S^{AB} = \tfrac{1}{4}[T^A, T^B],$$

with entries in $\{-1, 0, 1\}$; the S^{AB} generate Spin(4,4) and, with the reflections T^C, Pin(4,4); $\Gamma = T^0 \cdots T^7 = \mathrm{diag}(-I_8, I_8)$, $P_{L,R} = \frac{1}{2}(I_{16} \mp \Gamma)$, $\sigma_{16} = T^0 T^1 T^2 T^3$.

2. The field and its Lagrangian

dirac16complex $\psi = (\psi_0, \ldots, \psi_{15})$ has complex anticommuting (Grassmann) components and is a second-quantized fermion field; $\bar{\psi} = \psi^\dagger \sigma_{16}$.

$$\mathcal{L} = \tfrac{1}{2}\left(\bar{\psi} T^A \partial_A \psi - (\partial_A \bar{\psi}) T^A \psi\right) - m\bar{\psi}\psi - \frac{\lambda}{2}(\bar{\psi}\psi)^2, \qquad T^A \partial_A \psi = (m + \lambda \bar{\psi}\psi)\psi.$$

$\mathcal{L}$ is exactly real; the quadratic (mass) and quartic (self-coupling) terms are non-zero because the components are complex. Quantization with time x_4 gives $\{\psi_a, \psi_b^\dagger\} = K_{ab}\delta$, $K = -i\sigma_{16} T^4$, $K^2 = I$, with an (8,8) indefinite (Krein) metric, and an exact split into two mirror sectors of opposite energy: $\mathcal{L} = \mathcal{L}_D[\psi_+] - \mathcal{L}_D[\psi_-] - \frac{\lambda}{2}(S_+ - S_-)^2$ for fields independent of x_5, x_6, x_7.

3. Mirror of Pre-gravity6plane.nb and the 6-planes

Restricted to configurations that depend only on x_0 and x_4, the analogue of the notebook's $\Psi_{16}(x_0, x_4)$ ($m = -2M$), the 16 field equations split into the four blocks $\{0,5,8,13\}, \{1,4,9,12\}, \{2,7,10,15\}, \{3,6,11,14\}$, as in the old notebook; its exponential solutions $e^{pz+qt}u$ with $(6p+q)(6p-q) = m^2/H^2$, $z = 6Hx_0$, $t = Hx_4$, are exactly the Maple solutions of the old notebook. With the triality frames and vector forms $V_1 = \eta(\psi_1^T \sigma F_A{}^a)$, $V_2 = F_a{}^A \psi_2$, a Universe defines the 6-plane

$$M^6 = \{Z \in \mathbb{R}^8 : n_1 \cdot Z = 0,\ n_2 \cdot Z = 0\}, \qquad n_k = V_k[\mathrm{Re}\,\psi],$$

with $\binom{8}{2} = 28$ parameterizations; a pair of universes of energies $\pm E$ creates the 6-plane $\{Z : \mathrm{Re}V_2 \cdot Z = \mathrm{Im}V_2 \cdot Z = 0\}$ by itself.

4. Gravity, Kohn-Sham DFT, pair creation, Einstein-Rosen bridges

Three gravitational fields (diagonal vielbeins): primordial $\mathrm{diag}(\tan 6Hx_0, e^{-Hx_4}\sin^{-1/6} 6Hx_0\,(\times 3), 1, e^{Hx_4}\sin^{-1/6} 6Hx_0\,(\times 3))$ (the vielbein of Pre-gravity6plane.nb with $a_4 = Hx_4$); modern $\mathrm{diag}(f^{-1/2}, 1, 1, 1, f^{1/2}, 1, 1, 1)$ with $f = 1 - r_s/x_0$; future the same with $f = 1 - r_s/x_0 - H_\Lambda^2 x_0^2$. In each, the Euler-Lagrange equations derived from the Grassmann Lagrangian are $\gamma^\mu D_\mu \psi = (m + \lambda\bar{\psi}\psi)\psi$, e.g. (primordial)

$$\cot(6Hx_0) T^0 \partial_0 \psi + e^{Hx_4} \sin^{\frac{1}{6}}(6Hx_0) \sum_{i=1}^{3} T^i \partial_i \psi + T^4 \partial_4 \psi + e^{-Hx_4} \sin^{\frac{1}{6}}(6Hx_0) \sum_{j=5}^{7} T^j \partial_j \psi - 3H \cot^2(6Hx_0) T^0 \psi = (m + \lambda\bar{\psi}\psi)\psi.$$

The quartic term is treated in a Kohn-Sham (DFT) approximation, $M = m+\lambda\rho_s$, with the Kohn-Sham Hamiltonian $h_{KS} = \alpha p + BM\beta$ ($\alpha = -T^4T^0$, $\beta = -iT^4$) in tortoise coordinates, ground and first-excited (ΔSCF) states, and the finite-temperature thermodynamic effective potential $\Omega(T)$. Proven in each field: $\mathcal{L}_m[\Gamma\psi^*] = \mathcal{L}_{-m}[\psi]$ and $\mathcal{L}_m[T^1\psi(R_1x)] = \mathcal{L}_{-m}[\psi]$ (mass-sign pairing); $T^0h_{KS}T^0 = -h_{KS}$ and $h^*_{KS} = -h_{KS}$ (every level ϵ has a partner $-\epsilon$); in the zero-energy vacuum the creation of a + universe pair together with a − universe pair changes the energy by exactly 0, while single quanta and single-sign creation are forbidden by the conserved charges and energy: universes of masses $\{+m, -m\}$ are created in pairs. The 1935 Einstein-Rosen bridges are verified; the modern field has one and the future field two exact bridges (two congruent sheets, $g = 0$ on the bridge, finite curvature); the primordial field's degenerate hypersurface $x_0 = 0$ is a curvature singularity ($R \approx -\frac{31}{216H^2x_0^4}$); across a bridge the vacuum pairs universes of energy $+\epsilon$ and $-\epsilon$ with weight $1/(e^{\epsilon/T_H}+1)$, $T_H = \kappa/2\pi$.

5. Assumptions and interpretations made in this work

1. The modern and future gravitational fields (not specified in the request) are defined as above; the primordial field is the notebook's vielbein with $a_4(Hx_4) = Hx_4$.
2. "Universe" means a mode function (c-number solution) of dirac16complex, as the notebook calls Ψ_{16} the wave function of a Universe; its "mass" is its energy for translations of x_4 (or the mass parameter m for the mass-sign theorem).
3. The theorems are exact statements about this model; they are not a proof about the physical universe.

6. Reproducing everything

```
cd code
wolframscript -file verify_dirac_matrices.wls ../Pre-gravity6plane.nb
wolframscript -file build_nb2.wls
python3 gen_dm.py
python3 md2tex.py <file>.md <file>.tex ; pdflatex <file>.tex
```

7. Repository inventory

file	content
Pre-gravity6plane.nb	original notebook (P. L. Nash, GPL)
6planes.nb	new notebook, evaluated, 34 input cells, no messages
README.md/.tex/.pdf	this overview
dirac matrices.md/.tex/.pdf	proofs and display of the eight real 16x16 Dirac matrices
dirac16complex_6planes.md/.tex/.pdf	the field, Lagrangian, quantization, mirror of the old notebook, 6-planes
gravity_DFT_pair_creation.md/.tex/.pdf	three gravitational fields, Euler-Lagrange equations, Kohn-Sham DFT, pair theorems
einstein_rosen_bridges.md/.tex/.pdf	Einstein-Rosen bridges
*_provenance.md/.tex/.pdf	sources and computations behind each document
*_symbols.md/.tex/.pdf	definitions of all symbols of each document
code/	core.wl, verify_dirac_matrices.wls (+ log), dm_data.json, gen_dm.py, md2tex.py, 6planes_src.txt, build_nb2.wls (+ log)